

Part A — Basic grep

Exercise 1 — Search File Contents

Using payload.sh:

Find the line containing wget.

Find the line containing nc.

Find the line containing chmod.

Search for WGET ignoring case.

Display the line numbers containing /tmp.

Count the number of lines containing /tmp.

Display only the matched word wget.

Search for nc as a whole word.

```
(prathiksha@kali)~[/grep_lab]
$ grep 'wget' payload.sh
wget http://example.com/file

(prathiksha@kali)~[/grep_lab]
$ grep 'nc' payload.sh
nc 192.168.1.10 4444

(prathiksha@kali)~[/grep_lab]
$ grep 'chmod' payload.sh
chmod +x /tmp/test

(prathiksha@kali)~[/grep_lab]
$ grep -i 'WGET' payload.sh
wget http://example.com/file

(prathiksha@kali)~[/grep_lab]
$ grep -n '/tmp' payload.sh
5:echo "Downloading file to /tmp"
6:cp file /tmp/test
8:chmod +x /tmp/test

(prathiksha@kali)~[/grep_lab]
$ grep -c '/tmp' payload.sh
3

(prathiksha@kali)~[/grep_lab]
$ grep -o 'wget' payload.sh
wget

(prathiksha@kali)~[/grep_lab]
$ grep -w 'nc' payload.sh
nc 192.168.1.10 4444
```

Exercise 2 — Compare Files

Search for wget in:

payload.sh

update.sh

cleanup.sh

Answer:

Which file contains wget?

Which files do not contain wget?

Repeat the search for /tmp.

Which files contain /tmp?

Use grep with multiple filenames.

```
(prathiksha@kali)~[/grep_lab]
$ grep 'wget' payload.sh update.sh cleanup.sh
payload.sh:wget http://example.com/file
update.sh:wget http://example.com/update

(prathiksha@kali)~[/grep_lab]
$ grep -l 'wget' payload.sh update.sh cleanup.sh
payload.sh
update.sh

(prathiksha@kali)~[/grep_lab]
$ grep -L 'wget' payload.sh update.sh cleanup.sh
cleanup.sh

(prathiksha@kali)~[/grep_lab]
$ grep '/tmp' payload.sh update.sh cleanup.sh
payload.sh:echo "Downloading file to /tmp"
payload.sh:cp file /tmp/test
payload.sh:chmod +x /tmp/test
cleanup.sh:rm -f /tmp/test

(prathiksha@kali)~[/grep_lab]
$ grep -l '/tmp' payload.sh update.sh cleanup.sh
payload.sh
cleanup.sh

(prathiksha@kali)~[/grep_lab]
$ grep 'wget' payload.sh update.sh cleanup.sh
payload.sh:wget http://example.com/file
update.sh:wget http://example.com/update
```

Exercise 3 — Inverse Search

Using system.log:

Display all lines that do not contain INFO.

Display all lines that do not contain ERROR.

Display all lines except those containing WARNING.

Count the number of non-INFO entries.

```
(prathiksha@kali)~/grep_lab
$ grep -v 'INFO' system.log
WARNING Multiple login attempts detected
ERROR Failed to connect to database
WARNING Disk space is getting low
ERROR Database connection failed
FAILED Authentication attempt
ERROR Malware detection alert
WARNING Unusual network activity detected

(prathiksha@kali)~/grep_lab
$ grep -v 'ERROR' system.log
INFO System started
INFO User login successful
WARNING Multiple login attempts detected
INFO Network connection established
INFO Retrying database connection
WARNING Disk space is getting low
FAILED Authentication attempt
INFO System monitoring active
WARNING Unusual network activity detected
INFO Security scan completed

(prathiksha@kali)~/grep_lab
$ grep -v 'WARNING' system.log
INFO System started
INFO User login successful
INFO Network connection established
ERROR Failed to connect to database
INFO Retrying database connection
ERROR Database connection failed
FAILED Authentication attempt
INFO System monitoring active
ERROR Malware detection alert
INFO Security scan completed

(prathiksha@kali)~/grep_lab
$ grep -vc 'INFO' system.log
7
```

Part B — Security Log Analysis

Exercise 4 — Analyze system.log

Using grep:

Find all ERROR entries.

Find all WARNING entries.

Find the FAILED entry.

Count the number of ERROR entries.

Display line numbers containing WARNING.

Search for error irrespective of case.

Determine whether the word Malware occurs in the file without displaying output.

```
(prathiksha@kali)~/grep_lab
$ grep 'ERROR' system.log
ERROR Failed to connect to database
ERROR Database connection failed
ERROR Malware detection alert

(prathiksha@kali)~/grep_lab
$ grep 'WARNING' system.log
WARNING Multiple login attempts detected
WARNING Disk space is getting low
WARNING Unusual network activity detected

(prathiksha@kali)~/grep_lab
$ grep 'FAILED' system.log
FAILED Authentication attempt

(prathiksha@kali)~/grep_lab
$ grep -c 'ERROR' system.log
3

(prathiksha@kali)~/grep_lab
$ grep -n 'WARNING' system.log
3:WARNING Multiple login attempts detected
7:WARNING Disk space is getting low
12:WARNING Unusual network activity detected

(prathiksha@kali)~/grep_lab
$ grep -i 'error' system.log
ERROR Failed to connect to database
ERROR Database connection failed
ERROR Malware detection alert

(prathiksha@kali)~/grep_lab
$ grep -q 'Malware' system.log; echo $?
0
```

Exercise 5 — Context Investigation

An ERROR event has been detected in system.log.

Display two lines after each ERROR.

Display two lines before each ERROR.

Display one line before and one line after each ERROR.

Display two lines around FAILED.

Why can examining events before and after an error be useful during incident investigation?

```
(prathiksha@kali)~/grep_lab
$ grep -A 2 'ERROR' system.log
ERROR Failed to connect to database
INFO Retrying database connection
WARNING Disk space is getting low
ERROR Database connection failed
FAILED Authentication attempt
INFO System monitoring active
ERROR Malware detection alert
WARNING Unusual network activity detected
INFO Security scan completed

(prathiksha@kali)~/grep_lab
$ grep -B 2 'ERROR' system.log
WARNING Multiple login attempts detected
INFO Network connection established
ERROR Failed to connect to database
INFO Retrying database connection
WARNING Disk space is getting low
ERROR Database connection failed
FAILED Authentication attempt
INFO System monitoring active
ERROR Malware detection alert

(prathiksha@kali)~/grep_lab
$ grep -B 1 -A 1 'ERROR' system.log
INFO Network connection established
ERROR Failed to connect to database
INFO Retrying database connection
WARNING Disk space is getting low
ERROR Database connection failed
FAILED Authentication attempt
INFO System monitoring active
ERROR Malware detection alert
WARNING Unusual network activity detected

(prathiksha@kali)~/grep_lab
$ grep -C 2 'FAILED' system.log
WARNING Disk space is getting low
ERROR Database connection failed
FAILED Authentication attempt
INFO System monitoring active
ERROR Malware detection alert
```

Looking at the events surrounding an error helps reconstruct the full sequence of an incident — the lines before often reveal what triggered the failure (e.g. a suspicious login attempt or unusual activity), while the lines after show the consequence or response (e.g. a retry, an alert, or further compromise). Without that context, an isolated ERROR line only tells you that something failed, not why or what happened as a result.

Part C — egrep / grep -E

Exercise 6 — Search Multiple Security Events

Using system.log, write a single command to find lines containing:

ERROR

WARNING

FAILED

```
(prathiksha@kali)~/grep_lab
$ grep -E 'ERROR|WARNING|FAILED' system.log
WARNING Multiple login attempts detected
ERROR Failed to connect to database
WARNING Disk space is getting low
ERROR Database connection failed
FAILED Authentication attempt
ERROR Malware detection alert
WARNING Unusual network activity detected
```

Exercise 7 — Multiple Suspicious Commands

Using payload.sh, search for any of:

wget

curl

nc

chmod

using a single extended regular expression.

Answer:

Which patterns were found?

wget, nc, chmod

Which pattern was not found?

curl

How many lines contain at least one of these patterns?

```
(prathiksha@kali)-[~/grep_lab]
$ grep -E 'wget|curl|nc|chmod' payload.sh
wget http://example.com/file
nc 192.168.1.10 4444
chmod +x /tmp/test

(prathiksha@kali)-[~/grep_lab]
$ grep -Ec 'wget|curl|nc|chmod' payload.sh
3
```

Exercise 8 — Search Multiple Files

Which files contain at least one selected command?

Display only the filenames containing matches.

Display matching lines with line numbers.

Count the matching lines in each file.

```
(prathiksha@kali)-[~/grep_lab]
$ grep -El 'wget|curl|nc|chmod' payload.sh update.sh cleanup.sh
payload.sh
update.sh

(prathiksha@kali)-[~/grep_lab]
$ grep -En 'wget|curl|nc|chmod' payload.sh update.sh cleanup.sh
payload.sh:4:wget http://example.com/file
payload.sh:7:nc 192.168.1.10 4444
payload.sh:8:chmod +x /tmp/test
update.sh:5:wget http://example.com/update

(prathiksha@kali)-[~/grep_lab]
$ grep -Ec 'wget|curl|nc|chmod' payload.sh update.sh cleanup.sh
payload.sh:3
update.sh:1
cleanup.sh:0
```

Part D — patterns.txt

Your dataset already contains patterns.txt.

Exercise 9 — Pattern File Investigation

Use patterns.txt to:

Search payload.sh.

Search system.log.

Search config.conf.

Search all .sh files.

Display only filenames containing at least one pattern from patterns.txt.

Display matching lines without printing filenames.

Which files contain indicators listed in patterns.txt?

```
(prathiksha@kali)-[~/grep_lab]
$ grep -f patterns.txt payload.sh
wget http://example.com/file
nc 192.168.1.10 4444
chmod +x /tmp/test

(prathiksha@kali)-[~/grep_lab]
$ grep -f patterns.txt system.log
FAILED Authentication attempt
ERROR Malware detection alert

(prathiksha@kali)-[~/grep_lab]
$ grep -f patterns.txt config.conf
password=admin123

(prathiksha@kali)-[~/grep_lab]
$ grep -f patterns.txt *.sh
payload.sh:wget http://example.com/file
payload.sh:nc 192.168.1.10 4444
payload.sh:chmod +x /tmp/test
update.sh:wget http://example.com/update

(prathiksha@kali)-[~/grep_lab]
$ grep -lf patterns.txt *.sh system.log config.conf
payload.sh
update.sh
system.log
config.conf

(prathiksha@kali)-[~/grep_lab]
$ grep -hvf patterns.txt *.sh system.log config.conf
wget http://example.com/file
nc 192.168.1.10 4444
chmod +x /tmp/test
wget http://example.com/update
FAILED Authentication attempt
ERROR Malware detection alert
password=admin123
```

The files payload.sh and system.log contain indicators listed in patterns.txt - payload.sh matches wget, nc, and chmod, while system.log matches ERROR and FAILED — whereas config.conf does not contain any of the listed indicators.

Part E — Regex Using filelist.txt

Exercise 10 — Beginning and End of Line

Using filelist.txt:

Find filenames beginning with malware.

Find filenames beginning with script.

Find filenames ending in .sh.

Find filenames ending in .py.

Find filenames ending in .log.

Find filenames beginning with p and ending in .sh

```
(prathiksha@kali)~/grep_lab
$ grep '^malware' filelist.txt
malware1.sh
malware2.sh
malware3.py

(prathiksha@kali)~/grep_lab
$ grep '^script' filelist.txt
script.sh
script.py

(prathiksha@kali)~/grep_lab
$ grep '\.sh$' filelist.txt
malware1.sh
malware2.sh
script.sh
payload.sh
cleanup.sh
update.sh
user.sh

(prathiksha@kali)~/grep_lab
$ grep '\.py$' filelist.txt
malware3.py
script.py
scanner.py

(prathiksha@kali)~/grep_lab
$ grep '\.log$' filelist.txt
system.log
backup.log

(prathiksha@kali)~/grep_lab
$ grep -E '^p.*\.sh$' filelist.txt
payload.sh

(prathiksha@kali)~/grep_lab
$ grep -E '^p.*\.sh$' filelist.txt
payload.sh
```

Exercise 11 — Character Classes

Using filelist.txt:

Find malware1.sh and malware2.sh using one pattern.

Find filenames containing a digit.

Find filenames containing digits 1 or 2.

Find filenames beginning with m or p.

Find filenames beginning with s or u.

```
(prathiksha@kali)~/grep_lab
$ grep -E 'malware[12]\.sh' filelist.txt
malware1.sh
malware2.sh

(prathiksha@kali)~/grep_lab
$ grep '[0-9]' filelist.txt
malware1.sh
malware2.sh
malware3.py

(prathiksha@kali)~/grep_lab
$ grep '[12]' filelist.txt
malware1.sh
malware2.sh

(prathiksha@kali)~/grep_lab
$ grep '^([mp])' filelist.txt
malware1.sh
malware2.sh
malware3.py
payload.sh
program.pl
photo.jpg

(prathiksha@kali)~/grep_lab
$ grep '^([su])' filelist.txt
script.sh
script.py
scanner.py
update.sh
system.log
user.sh
```

Exercise 12 — Negated Character Classes

Using filelist.txt:

Find filenames that do not begin with m.

Find filenames that do not begin with m or p.

Find malware filenames where the character following malware is not a digit.

```
(prathiksha@kali)~/grep_lab
$ grep '^m' filelist.txt
crpt.sh
crpt.py
payload.sh
canner.py
cleanup.sh
update.sh
eport.txt
system.log
rogram.pl
hoto.jpg
ser.sh
ackup.log

(prathiksha@kali)~/grep_lab
$ grep -E '[^mp]' filelist.txt
crpt.sh
crpt.py
canner.py
cleanup.sh
update.sh
eport.txt
system.log
ser.sh
ackup.log

(prathiksha@kali)~/grep_lab
$ grep -E 'malware[^0-9]' filelist.txt
```

Exercise 13 —

Write a shell script that accepts a filename and determines whether it exists.

Test with:

payload.sh

report.txt

unknown.sh

Write a script that accepts a path and determines whether it is:

a regular file,

a directory,

or neither.

Test with:

payload.sh

system.log

quarantine

backup

Unknown

```
(prathiksha@kali)~/grep_lab
$ nano exists.sh

(prathiksha@kali)~/grep_lab
$ chmod +x exists.sh

(prathiksha@kali)~/grep_lab
$ ./exists.sh payload.sh
File exists

(prathiksha@kali)~/grep_lab
$ ./exists.sh report.txt
File exists

(prathiksha@kali)~/grep_lab
$ ./exists.sh unknown.sh
File does not exist

(prathiksha@kali)~/grep_lab
$ nano filetype.sh

(prathiksha@kali)~/grep_lab
$ chmod +x filetype.sh

(prathiksha@kali)~/grep_lab
$ ./filetype.sh payload.sh
payload.sh is a regular file

(prathiksha@kali)~/grep_lab
$ ./filetype.sh system.log
system.log is a regular file

(prathiksha@kali)~/grep_lab
$ ./filetype.sh quarantine
quarantine is a directory

(prathiksha@kali)~/grep_lab
$ ./filetype.sh backup
backup is a directory

(prathiksha@kali)~/grep_lab
$ ./filetype.sh unknown
unknown is neither a regular file nor a directory
```

Exercise 15

Write a script that reports whether a selected file is:

readable,

writable,

executable.

Test with:

payload.sh

update.sh

report.txt

config.conf

Expected style of output:

File: payload.sh

Readable : Yes

Writable : Yes

Executable : Yes

```
(prathiksha@kali)~/grep_lab
$ nano permcheck.sh
(prathiksha@kali)~/grep_lab
$ chmod +x permcheck.sh
./permcheck.sh payload.sh
./permcheck.sh update.sh
./permcheck.sh report.txt
./permcheck.sh config.conf
File: payload.sh
Readable : Yes
Writable : Yes
Executable : Yes
File: update.sh
Readable : Yes
Writable : Yes
Executable : Yes
File: report.txt
Readable : Yes
Writable : Yes
Executable : No
File: config.conf
Readable : Yes
Writable : Yes
Executable : No
```

Exercise 16 —

Compare:

payload.sh

empty.txt

Write a script that determines whether each file contains data.

Display:

File contains data

or:

File is empty

```
(prathiksha@kali)~/grep_lab
$ nano hasdata.sh
(prathiksha@kali)~/grep_lab
$ chmod +x hasdata.sh
./hasdata.sh payload.sh empty.txt
payload.sh: File contains data
empty.txt: File is empty
```

Part G

Exercise 17 — File Validation Using []

Write a script that accepts a filename and checks:

Does it exist?

Is it a regular file?

Is it readable?

Is it non-empty?

```
(prathiksha@kali)~/grep_lab
$ nano validate.sh
(prathiksha@kali)~/grep_lab
$ chmod +x validate.sh
./validate.sh payload.sh
Exists : Yes
Regular file : Yes
Readable : Yes
Non-empty : Yes
```

Exercise 18 — [] + grep

Write a script with the following logic:

Does file exist?

↓

Is it a regular file?

↓

Is it readable?

↓

Is it non-empty?

↓

Search for "wget"

If wget is detected, display:

Selected indicator detected – investigate further

Otherwise:

Selected indicator not detected

Test using both:

payload.sh

[update.sh](#)

```
(prathiksha@kali)-[~/grep_lab]
$ nano validate_grep.sh

(prathiksha@kali)-[~/grep_lab]
$ chmod +x validate_grep.sh
./validate_grep.sh payload.sh
./validate_grep.sh update.sh
Selected indicator detected – investigate further
Selected indicator detected – investigate further
```

Part H

Exercise 19 — Repeat File Tests Using [[]]

Repeat Exercise 17 using:

```
(prathiksha@kali)-[~/grep_lab]
$ nano validate2.sh

(prathiksha@kali)-[~/grep_lab]
$ chmod +x validate2.sh
./validate2.sh payload.sh
Exists      : Yes
Regular file : Yes
Readable    : Yes
Non-empty   : Yes
```

Exercise 20 — Regex Using [[]]

Write a shell script that accepts a filename and checks whether the filename ends with .sh.

Use:

[["\$filename" =~ regex]]

Display:

Shell script identified for inspection

or:

File does not have a .sh extension

Test:

payload.sh

scanner.py

Report.txt

```
(prathiksha@kali)-[~/grep_lab]
$ nano isshell.sh

(prathiksha@kali)-[~/grep_lab]
$ chmod +x isshell.sh
./isshell.sh payload.sh
./isshell.sh scanner.py
./isshell.sh report.txt
Shell script identified for inspection
File does not have a .sh extension
File does not have a .sh extension
```

Exercise 21 — Multiple Extensions Using [[]]

Extend the previous script to identify:

.sh

.py

.pl

files.

Test:

payload.sh

scanner.py

report.txt

config.conf

Do not classify these as malicious. Display:

Script file – inspect if required

```
(prathiksha@kali)-[~/grep_lab]
$ nano isscript.sh

(prathiksha@kali)-[~/grep_lab]
$ chmod +x isscript.sh
./isscript.sh payload.sh
./isscript.sh scanner.py
./isscript.sh report.txt
./isscript.sh config.conf
Script file – inspect if required
Script file – inspect if required
Not a recognized script file
Not a recognized script file
```

Part I — grep + File Operators + Constructs

Exercise 22 — Executable File Content Analysis

Write a script that:

Accepts a filename.

Checks whether it exists.

Checks whether it is a regular file.

Checks whether it is executable.

If executable, searches for wget

Test with:

payload.sh

update.sh

report.txt

```
(prathiksha@kali)-[~/grep_lab]
$ nano exec_scan.sh

(prathiksha@kali)-[~/grep_lab]
$ chmod +x exec_scan.sh
./exec_scan.sh payload.sh
./exec_scan.sh update.sh
./exec_scan.sh report.txt
payload.sh is executable – scanning content
wget found in payload.sh
update.sh is executable – scanning content
wget found in update.sh
report.txt exists but is not executable – skipping content scan
```

Exercise 23 — egrep + File Operators

Write a script that performs content analysis only when the selected file is:

a regular file,

readable,

non-empty.

If the checks succeed, use egrep/grep -E to search for:

wget|curl|nc|chmod

Display:

One or more selected indicators detected

if a match occurs.

Otherwise:

Selected indicators not detected

```

(prathiksha@kali)-[~/grep_lab]
└─$ nano indicator_scan.sh

(prathiksha@kali)-[~/grep_lab]
└─$ chmod +x indicator_scan.sh
./indicator_scan.sh payload.sh
One or more selected indicators detected

(prathiksha@kali)-[~/grep_lab]
└─$ ./indicator_scan.sh report.txt # likely "Selected indicators not detected"
./indicator_scan.sh empty.txt # "Pre-checks failed" - fails the -s test
./indicator_scan.sh unknown.sh # "Pre-checks failed" - fails the -f test
Selected indicators not detected
Pre-checks failed - file not a regular, readable, non-empty file
Pre-checks failed - file not a regular, readable, non-empty file

```

Part J — Configuration File Investigation

Exercise 24 — Sensitive Information Search

Use config.conf.

First verify that it:

exists,

is a regular file,

is readable,

is non-empty.

Then use egrep to search for:

password|username

Display the matching lines.

Why should configuration files containing authentication information receive additional protection?

```

(prathiksha@kali)-[~/grep_lab]
└─$ nano config_check.sh

(prathiksha@kali)-[~/grep_lab]
└─$ chmod +x config_check.sh
./config_check.sh
Pre-checks passed - searching for sensitive keywords

username=admin
password=admin123

```

Part K

Exercise 25 — Suspicious File Triage

Write a Bash program called:

file_check.sh

The program should ask the user to enter a filename.

It should:

Determine whether the path exists.

Determine whether it is a regular file.

Determine whether it is a directory.

If it is a regular file, determine whether it is readable.

Determine whether it is writable.

Determine whether it is executable.

Determine whether it is empty/non-empty.

If readable and non-empty, use grep to search for wget.

Use egrep/grep -E to search for wget|curl|nc|chmod.

Use \$? to determine whether a selected search succeeded.

Use [[]] with regex to determine whether the filename ends in .sh.

Display a summary of the findings.

Example:

Enter filename: payload.sh

----- FILE INVESTIGATION -----

Exists : Yes

Regular File : Yes

Directory : No

Readable : Yes

Writable : Yes

Executable : Yes

Non-empty : Yes

Shell Script : Yes

wget : Found
Selected indicators : Found
Further inspection recommended

```
(prathiksha@kali)-[~/grep_lab]
$ nano file_check.sh

(prathiksha@kali)-[~/grep_lab]
$ chmod +x file_check.sh
./file_check.sh
Enter filename: payload.sh

——— FILE INVESTIGATION ———

Exists      : Yes
Regular File : Yes
Directory   : No
Readable    : Yes
Writable    : Yes
Executable  : Yes
Non-empty   : Yes
Shell Script : Yes

wget        : Found
Selected indicators : Found

Further inspection recommended

(prathiksha@kali)-[~/grep_lab]
$ ./file_check.sh
Enter filename: report.txt

——— FILE INVESTIGATION ———

Exists      : Yes
Regular File : Yes
Directory   : No
Readable    : Yes
Writable    : Yes
Executable  : No
Non-empty   : Yes
Shell Script : No

wget        : Not Found
Selected indicators : Not Found

No indicators detected
```

